

Analysis of Scientific Data



Lecture 11

Categorical Data Analysis

Module 9 – Cognitive Failures

A broad epidemiological study included the following question to measure cognitive failures experienced by an individual: ‘Have you experienced problems of memory (e.g. forgetting where you put things), attention (e.g. failures of concentration) or action (doing the wrong thing)?’ Responses were combined into a single binary response for each person – “yes” (experienced some type of cognitive failure) or “no” (experienced no cognitive failure).

One group of researchers using the data from this study were interested in whether caffeine consumption (mg/day) had an effect on cognitive failure.

Data based on Andrew P Smith (2008) Caffeine, cognitive failures and health in a non-working community sample. *Human Psychopharmacology* 24, 29-34.

Poll Question

What are the log odds of getting heads on a single toss of a fair coin?

$$\ln \left(\frac{p}{1-p} \right)$$

$$\ln \left(\frac{0.5}{1-0.5} \right) = \ln(1) = 0$$

1. -1

2. 0

3. 0.5

4. 1

Poll Question

For the logistic regression model

$$\ln \left(\frac{p}{1-p} \right) = b_0 + b_1 x,$$
$$b_0 + b_1 \left(-\frac{b_0}{b_1} \right) = 0$$

what value of x gives an even chance of the response?

$$p = \frac{1}{2}$$

1. b_1/b_0
2. $-b_1/b_0$
3. b_0/b_1
4. $-b_0/b_1$

Logistic regression

Is there an association between cognitive failure and caffeine?

Test $H_0: \beta_i = 0$ vs $H_1: \beta_i \neq 0$

Test statistic $Z = \frac{b_1 - 0}{se(b_1)} = -2.434$

$p\text{-value} = 0.0149$

moderate evidence

Prediction for caffeine = 300

$$\begin{aligned}\log \text{ odds} &= \log\left(\frac{p}{1+p}\right) = 0.14628 - 0.003888 (\text{caffeine}) \\ &= \quad \quad \quad " \quad \quad \quad (300) \\ &= -1.02002\end{aligned}$$

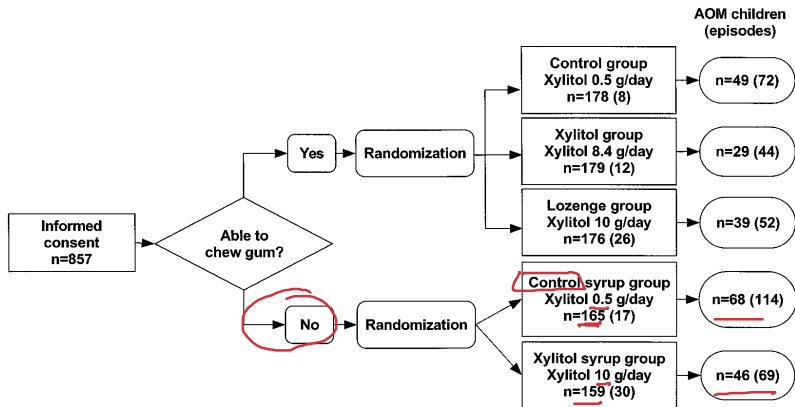
$$\text{odds} = e^{-1.02002} = 0.3605877$$

$$p = \frac{\text{odds}}{1 + \text{odds}} = \frac{0.3605877}{1.3605877} = 0.265$$

Preventing Ear Infections

A comparative experiment looked at the effectiveness of the sweetener xylitol in preventing ear infections in preschool children.

Experimental Design



Poll Question

Let X be the number of middle ear infections experienced by a random person during their childhood.

A suitable probability model for X would be

~~1. $X \sim \text{Binomial}(n, p)$~~

$X = 0, 1, 2, \dots, n$

2. $X \sim \text{Poisson}(\lambda)$

$\lambda = np, n \rightarrow \infty$

~~3. $X \sim \text{Normal}(\mu, \sigma)$~~

~~4. $X \sim T_{n-1}$~~

~~5. $X \sim \text{Uniform}(a, b)$~~

Poll Question

Let X be the number of children in the placebo group who experienced at least one middle ear infection during the study.

A suitable probability model for X would be

1. $X \sim \text{Binomial}(n, p)$

2. $X \sim \text{Poisson}(\lambda)$

3. $X \sim \text{Normal}(\mu, \sigma)$

4. $X \sim T_{n-1}$

5. $X \sim \text{Uniform}(a, b)$

Ear Infection Data

The results from the 'syrup' groups can be summarised in a two-way table of counts:

Syrup	Infection	No Infection	Total
Placebo	<u>68</u>	97	<u>165</u>
Xylitol	46	113	159

Is there any evidence that Xylitol reduces the risk of middle ear infection in children?

Uhari, M., Kenteerkar, T. and Niemala, M. (1998) "A novel use of Xylitol sugar in preventing acute otitis media". *Pediatrics*, **102**, 879–884. Appearing in Utts, J.M. and Heckard, R.F. (2007) *Mind on Statistics* (3rd edition).

Let p_1 = probability of ear infection with placebo
 p_2 = " " " " " xylitol

Test $H_0: p_1 = p_2$ vs $H_1: p_1 > p_2$
or $p_1 - p_2 = 0$ or $p_1 - p_2 > 0$

Observed $\hat{p}_1 = \frac{68}{165} = 0.41212$ and $\hat{p}_2 = \frac{46}{159} = 0.2893$

so $\hat{p}_1 - \hat{p}_2 = \underline{0.1228}$

$$\begin{aligned}\text{Test statistic } Z &= \frac{\text{estimate} - \text{hypothesized}}{\text{SE}(\text{estimate})} \\ &= \frac{0.1228 - 0}{0.05255} \\ &= 2.337\end{aligned}$$

p-value $P(p_1 - p_2 > 0.1228)$
 $= P(Z > 2.337)$
 $= \underline{0.0097}$, strong evidence that infection rate is higher with placebo

Confidence interval for $p_1 - p_2$
 $CI = \text{estimate} \pm Z^* \text{se}(\text{estimate})$
 $= 0.1228 \pm 1.96 \times 0.05255$
 $= 0.1228 \pm 0.103$
 $= (0.02, 0.23)$

$$\begin{aligned}\text{se}(\hat{p}_1 - \hat{p}_2) &= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \\ &= \sqrt{\frac{0.41212(1-0.41212)}{165} + \frac{0.2893(1-0.2893)}{159}} \\ &= 0.05255\end{aligned}$$

The Cochrane Library provides systematic reviews of clinical research, assessing the quality and results from numerous papers on the same topic.

For example, this is the current Cochrane Review on the effectiveness of xylitol on ear infections:

- Azarpazhooh A, Lawrence HP, Shah PS. Xylitol for preventing acute otitis media in children up to 12 years of age. *Cochrane Database of Systematic Reviews* 2016, Issue 8. Art. No.: CD007095. DOI: 10.1002/14651858.CD007095.pub3

Respiratory Syncytial Virus (RSV) Incidence

Queensland Health splits the state into 14 Hospital and Health Service (HHS) regions. The following table gives the number of RSV cases reported for the week ending 3 May 2026 for five of the HHS regions:

Gold Coast	Metro North	Metro South	Sunshine Coast	West Moreton
74	97	135	42	56

Is there a difference in RSV reports between these regions?

<https://www.health.qld.gov.au/clinical-practice/guidelines-procedures/diseases-infection/surveillance/reports/flu>

Populations

The populations of the five HHS regions are as follows:

Gold Coast	Metro North	Metro South	Sunshine Coast	West Moreton
663216	1084001	1236517	468233	325236

<https://www.choreport.health.qld.gov.au/our-people/demography>

	GC	MN	MS	SC	WM
Observed	74	97	135	42	56
Expected	70.94	115.94	132.25	50.08	34.79

H_0 : all regions have the same RSV incidence rate

$$\text{expected rate} = \frac{\text{total cases}}{\text{total population}} = \frac{74 + 97 + \dots + 56}{663216 + \dots + 325236} = 0.0001$$

$$\text{expected cases for GC} = 0.0001 \times 663216 = 70.94$$

$$\chi^2 = \frac{(74 - 70.94)^2}{70.94} + \frac{(97 - 115.94)^2}{115.94} + \dots + \frac{(56 - 34.79)^2}{34.79}$$

$$= 17.52$$

$$\begin{aligned} p\text{-value} &= P(\chi^2_{df} \geq 17.52) \\ &= P(\chi^2_4 \geq 17.52) \\ &= 0.0015 \end{aligned}$$

Strong evidence that the RSV incidence rate is not the same across the regions.

cases = c(74, 97, 135, 42, 56)

population = c(663216, 1084001, 1236517, 468233, 325236)

chisq.test(cases, p=population, rescale.p=TRUE)

Test of Independence

Revisit the ear infection data:

Syrup	Infection	No Infection	Total
Placebo	68 (58.06)	97 (106.95)	165
Xylitol	46 (55.94)	113 (103.06)	159
total	114	210	324

Is there any evidence that Xylitol is associated with ear infection?

H_0 : no association between Xylitol and infection

H_1 : there is an association between Xylitol and infection

If H_0 is true: Xylitol is independent of infection

$$\begin{aligned} P(\text{placebo, infection}) &= P(\text{placebo}) P(\text{infection}) \\ &= \frac{165}{324} \times \frac{114}{324} \end{aligned}$$

Expected cases for placebo, infection = $\left[\frac{165 \times 114}{324} \right] \times 324 = \frac{165 \times 114}{324} = 58.06$

row total column total

total cases

$$\text{Test statistic } \chi^2 = \frac{(68 - 58.06)^2}{58.06} + \frac{(46 - 55.44)^2}{55.44} + \frac{(97 - 106.95)^2}{106.95} + \frac{(113 - 103.06)^2}{103.06}$$

$$= 5.35$$

p-value is $P(\chi^2_{(1)} \geq 5.35)$

$$= 0.021$$

moderate evidence that Xylitol is not independent of infection

10	46
97	113

✓	✓
✓	✓

$$df = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$$

```
dat = matrix(c(68, 46, 97, 113), nrow=2)
```

```
chisq.test(tab, correct=FALSE)
```